

April 10, 2012

Silicon Errata for EZ-USB® FX2LP™ CY7C68013A/14A/15A/16A Rev. B

This document describes the errata for the EZ-USB® FX2LP™ CY7C68013A/14A/15A/16A Rev. B silicon. Details include errata trigger conditions, scope of impact, available workarounds, and silicon revision applicability. Compare this document to the device's datasheet for a complete functional description.

Contact your local Cypress Sales Representative if you have questions.

Part Numbers Affected

Part Number	Package Type	Operating Range
CY7C68013A	All	Commercial
CY7C68014A	All	Commercial
CY7C68015A	All	Commercial
CY7C68016A	All	Commercial

CY7C68013A/14A/15A/16A Qualification Status

In production

CY7C68013A/14A/15A/16A Errata Summary

This table defines the errata for available CY7C68013A/14A/15A/16A family devices. An "X" indicates that the errata pertain to the selected device.

Items	CY7C68013A/14A/15A/16A	Silicon Revision	Fix Status
[1] Empty Flag Assertion	X	B	No silicon fix planned currently. Use workaround

1. Empty Flag Assertion

■ **PROBLEM DEFINITION**

In Slave FIFO Asynchronous Word Wide mode, if a single word data is transferred from the USB host to EP2, configured as OUT Endpoint (EP) in the first transaction, then the Empty flag behaves incorrectly. This does not happen if the data size is more than one word in the first transaction.

■ **PARAMETERS AFFECTED**

NA

■ **TRIGGER CONDITION(S)**

In Slave FIFO Asynchronous Word Wide Mode, after firmware boot and initialization, EP2 OUT endpoint empty flag indicates status as 'Empty'. When data is received in EP2, the status changes to 'Not-Empty'. However, if data transferred to EP2 is a single word, then asserting SLRD with FIFOADR pointing to any other endpoint changes 'Not-Empty' status to 'Empty' for EP2 even though there is a word data (or it is untouched). This is noticed only when the single word is sent as the first transaction and not if it follows a multi-word packet as the first transaction.

- **SCOPE OF IMPACT**

External interface does not see data available in EP2 OUT endpoint and can end up waiting for data to be read.

- **WORKAROUND**

One of the following workarounds can be used:

- Send a pulse signal to the SLWR pin, with FIFOADR pins pointing to an endpoint other than EP2, after firmware initialization and before or after transferring the data to EP2 from the host
- Set the length of the first data to EP2 to be more than a word
- Prioritize EP2 read from the Master for multiple OUT EPs and single word write to EP2
- Write to an IN EP, if any, from the Master before reading from other OUT EPs (other than EP2) from the Master.

- **FIX STATUS**

There is no silicon fix planned for this currently; use the workarounds provided.

References

[1] Document # 38-08032, CY7C68013A/CY7C68015A/CY7C68014A/CY7C68016A EZ-USB FX2LP USB Microcontroller High-Speed USB Peripheral Controller

Document History

Document Title: Silicon Errata for EZ-USB[®] FX2LP[™] CY7C68013A/14A/15A/16A Rev. B Document Number: 38-17037			
Rev.	ECN No.	Orig. of Change	Description of Change
**	338880	MON	New errata for rev B silicon
*A	3207724	ODC/DBIR	Added Errata item #1 related to Empty Flag Assertion
*B	3565494	GAYA	Modified Errata item #1

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